

Tugas KKA Week 14
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1. Buat FOL dari kalimat berikut:

a. Every gardener likes the sun

Predicates:

- Gardener(x) : x is a gardener
- Likes(x, y) : x likes y

Answer:

$\forall x(\text{Gardener}(x) \rightarrow \text{Likes}(x, \text{Sun}))$

b. All purple mushrooms are poisonous

Predicates:

- PurpleMushroom(x) : x is a purple mushroom
- Poisonous(x) : x is poisonous

Answer:

$\forall x(\text{PurpleMushroom}(x) \rightarrow \text{Poisonous}(x))$

c. You can fool some of the people all of the time

Predicates:

- Person(x) : x is a person
- Fool(x, y, t) : x fools y at time t

Answer:

$\exists y(\text{Person}(y) \wedge \forall t \text{Fool}(\text{You}, y, t))$

d. You can fool all of the people some of the time

Predicates:

- Person(x) : x is a person
- Fool(x, y, t) : x fools y at time t

Answer:

$\forall y(\text{Person}(y) \rightarrow \exists t \text{Fool}(\text{You}, y, t))$

e. No purple mushroom is poisonous

Predicates:

- PurpleMushroom(x) : x is a purple mushroom
- Poisonous(x) : x is poisonous

Answer:

$\forall x(\text{PurpleMushroom}(x) \rightarrow \neg \text{Poisonous}(x))$

2. Problem Statement:

Knowledge Base (KB):

- a. All cats like fish.
- b. Cats eat everything they like.
- c. Ziggy is a cat.

Goal Query:

Does Ziggy eat fish?

Tasks:

- a. Convert the KB into First-Order Logic (FOL).
- b. Solve the query using Forward Chaining.

- c. Solve the query using Backward Chaining.
- d. Solve the query using Resolution.

Answer:

- a. Convert the KB into First-Order Logic (FOL).

Predicates:

- $Cat(x)$: x is a cat
- $Likes(x,y)$: x likes y
- $Eats(x,y)$: x eats y

KB in FOL:

- a. All cats like fish: $\forall x(Cat(x) \rightarrow Likes(x, fish))$
- b. Cats eat everything they like: $\forall x \forall y((Cat(x) \wedge Likes(x,y)) \rightarrow Eats(x,y))$
- c. Ziggy is a cat: $Cat(Ziggy)$

- b. Solve the query using Forward Chaining.

Facts: $Cat(Ziggy)$

Rules:

- $Cat(x) \rightarrow Likes(x, fish)$
- $Cat(x) \wedge Likes(x, y) \rightarrow Eats(x, y)$

FC Steps:

1. From $Cat(Ziggy)$ and rule 1 $\rightarrow Likes(Ziggy, fish)$
2. From $Cat(Ziggy)$ and $Likes(Ziggy, fish)$ and rule 2 $\rightarrow Eats(Ziggy, fish)$

Conclusion: Ziggy eats fish.

- c. Solve the query using Backward Chaining.

Goal: $Eats(Ziggy, fish)$

BC Steps:

1. To prove $Eats(Ziggy, fish)$, use rule 2 $\rightarrow$ need $Cat(Ziggy)$ and $Likes(Ziggy, fish)$
2. $Cat(Ziggy)$ is a fact
3. To prove $Likes(Ziggy, fish)$, use rule 1 + $Cat(Ziggy) \rightarrow Likes(Ziggy, fish)$

Conclusion: Ziggy eats fish.

- d. Solve the query using Resolution.

CNF Clauses:

- $\neg Cat(x) \vee Likes(x, fish)$
- $\neg Cat(x) \vee \neg Likes(x, y) \vee Eats(x, y)$
- $Cat(Ziggy)$
- $\neg Eats(Ziggy, fish)$ (negated goal)

RES Steps:

1. (2) with $Ziggy, fish$ + (4) $\rightarrow \neg Cat(Ziggy) \vee \neg Likes(Ziggy, fish)$
2. Resolve with (1) for $Ziggy \rightarrow \neg Cat(Ziggy)$
3. Resolve with (3) $\rightarrow$ empty clause

Conclusion: Ziggy eats fish.